VxMark v4.1 Tests of Normal Function

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Defining “Normal Function”

This document describes essential procedures and tests to perform on VxMark v4.1 systems to confirm “normal” desired function. Key functions are listed below, corresponding to different tests.

Terms

- ATI = Audio-tactile interface; accessible controller
- EMC = electromagnetic compatibility
- HWTA = Hardware Test App
- MK = VxMark
- PAT = Personal Assistive Technology
- UPS = universal power supply

Key System Functions

The following functions should work normally both before and after any EMC, environmental, vibrational, and other mechanical tests on the system:

1. **Marks and prints ballots**. Ballot paper is between 8.5” x 11” to 14” paper, depending on the election tested, between 28-40lb weight, as supplied by VotingWorks. Counts accurate numbers of printed ballots.
2. **Visually displays** instructions and accurate feedback to users.
3. **Audibly plays** instructions and accurate feedback to users.
4. **Accepts user inputs on a touchscreen**.
5. **Accepts user inputs in the ATI.**
6. **Accepts inputs on the barcode scanner**.

Additional functions must also be confirmed to work normally before and after environmental and vibration tests, or any test that affects the corresponding mechanical parts. These include:

7. **Maintains physical security**, restricting administrative access and preventing tampering without evidence.
8. **Protects internal electronics and subsystems**. This includes configurations meant for normal operations and for storage.

Key System Components

The VxScan systems also have key components that should always be inspected visually for damage, if functional tests above do not already reveal the damage. These are listed below:

1. Computer
2. Touchscreen
3. Power Supply, internal
4. Smart Card Reader
5. USB-A receptacle
6. USB 3 Switch
7. Barcode Scanner
8. Cable Assemblies
9. Uninterruptible Power Supply (external to unit)
10. Printer (external to unit)

Additional components should be inspected for damage following environmental tests, vibration tests, other mechanical tests, or other stress tests that affect these parts:

11. Outer case
12. Inner panels around electronics
13. USB cover
14. Security tie points
15. Fasteners, both external and internal
16. Wires, ports, and electrical connections

Prepare the System for Testing

Prepare Software for Testing / Image the System

If needed, image the system with a test version of the software following the steps in the internal document *All Things Imaging*. Use an appropriate image defined in the internal document *Software Image Archive*. It can either be a production image or a Hardware Test App (HWTA) image.

Configure a test election according to the instructions in the User Manual, using an appropriate election definition and corresponding smart cards and USB sticks configured through a VxAdmin machine.

Prepare Hardware for Testing

Before doing the Tests of Normal Function described below, set up the VxMark (MK) unit along with a connected printer and UPS if possible. following these general steps:

1. Inspect the physical system components briefly. Note any existing concerns or problems before stress tests. Discuss, fix, or address them as needed. Some issues to watch out for include:
    a. Damage: Cracks, wear, holes, cuts, scratches, deformation, corrosion
    b. Dirtiness: Residues, stickiness, markings, dust, debris, discoloration, burns
    c. Looseness: Loose fasteners, loose cables, noises indicating loose parts
    d. Seizing: Frozen joints, stuck hinges, blocked openings, part interference

2. Open the polls according to the User Manual.

If you cannot set up the system due to faulty parts or system errors, that is a failure. Investigate the causes of this failure, and discuss with VotingWorks staff how to address it in order to continue tests, if possible.

Fully testing the unit should also include scanning the printed ballot on a VxScan unit and potentially reconfiguring the unit with a VxAdmin. Set up the VxScan and VxAdmin units nearby.

Pack up Hardware after Testing

After doing the Tests of Normal Function described below, pack up the hardware appropriately if needed for storage or transport. This includes removing smart cards and USB sticks from the system and stowing the headphones and cords. Store the hardware in a climate-controlled location until ready for the next batch of tests.

If you cannot close down and pack up the system due to faulty parts, that is a failure. Investigate the causes of this failure, and discuss with VotingWorks staff how to address it.

If the unit must be packaged for testing or shipping, then follow the standard packaging instructions in the internal document: *VxMark Production Playbook & Checklists: Packaging Checklist*.

Tests of Normal Function

After preparing the hardware for testing, perform the following tests of normal function. Record data appropriately, such as in the internal document: *VxMark v4.1 Tests of Normal Function - Checklist Template.*

Materials

Gather the following:

- VxMark unit
- Printer
- Printer paper:
  - 8.5” x 11” to 14”, between 28-43lb weight, white, any finish; good quality without tears or other disruptions; quantity: at least 2 sheets.
- (*as needed*) Printer toner
- UPS
- VxScan
- VxAdmin
- (*as needed*) Cleaning materials for VxMark unit
  - Microfiber cloth & 70% isopropyl alcohol
  - Compressed air / vacuum for electronics
- (*as needed*) Repair tools for VxMark unit
  - Hex wrench, magnets, light, etc

Required Tests

All of the following tests should be done before and after exposing the VxMark unit or subsystem to a stress test.

1. Marks and prints valid ballots:

a. Procedure:
   i. Mark and print at least 2 ballots.
   ii. At least one ballot should have a long write-in.

b. Acceptable results (PASS):
   i. Can mark ballots without errors.
   ii. Ballots print without errors.
   iii. Printed paper ink is clean, without smudges, streaks, holes, or fading.
      1. Check the toner and replace if needed; retest to confirm.
   iv. Ballot scans in VxScan without errors (assuming VxScan is clean and also has no issues).

c. Unacceptable results (FAIL):
   i. Software fails.
   ii. Cannot mark ballots, or difficult to mark ballots.
   iii. Ballots cannot print.
   iv. Ballots jam or become damaged.
   v. Ballot ink is smudged, has streaks, has holes, or is faded, when the toner has not reached its expected end of life.
   vi. Ballot is not scannable by VxScan.

    vii. Ballot is misinterpreted by VxScan, as checked in either the printed tally report or in VxAdmin.

d. Mixed results requiring discussion (MIXED):
    i. Poor ink quality when the toner was at end of life.
    ii. Printer is very loud or operates unusually.

2. Visually displays instructions and accurate feedback to users.

a. Procedure:
    i. During ballot marking and printing, analyze all button presses.
    ii. Change display settings and check for anomalies.
b. Acceptable results (PASS):
    i. Gives instructions for marking and printing a ballot.
c. Unacceptable results (FAIL):
    i. Inaccurate or missing feedback.
    ii. Screen failures.
d. Mixed results requiring discussion (MIXED):
    i. n/a

3. Audibly plays instructions and accurate feedback to users.

a. Procedure:
    i. Without headphones, during ballot marking and printing, analyze all sounds.
    ii. With headphones plugged in, during ballot marking and printing, analyze all sounds.
    iii. With headphones plugged in, change audio volume, rate, and language, and check for responsiveness.
b. Acceptable results (PASS):
    i. Gives instructions for marking and printing a ballot.
c. Unacceptable results (FAIL):
    i. On-screen feedback and audio feedback not in sync.
    ii. No sound; on mute.
    iii. Headphone failures.
d. Mixed results requiring discussion (MIXED):
    i. n/a

4. Accepts user inputs on the touchscreen.

a. Procedure:
    i. Alter the Settings with the buttons in the header.
b. Acceptable results (PASS):
    i. Screen responds appropriately to touch on buttons and selections.

        ii. Scrolling possible.
    c. Unacceptable results (FAIL):
        i. Touchscreen doesn’t respond.
        ii. Touchscreen is slow to respond or lags.
        iii. False touches.
    d. Mixed results requiring discussion (MIXED):
        i. n/a

5. **Accepts user inputs on the ATI**.
    a. Procedure:
        i. Use the ATI to alter settings and mark a ballot.
        ii. Plug in headphones and analyze the audio interface.
        iii. Pull the ATI out of its cradle and use it supported on something (e.g. hand, lap, table)
    b. Acceptable results (PASS):
        i. Navigation via ATI works as expected.
        ii. Audio control via ATI and headphones works as expected.
        iii. ATI remains stably connected.
    c. Unacceptable results (FAIL):
        i. ATI has wrong or unexpected responses.
        ii. Headphones do not function.
        iii. ATI disconnects or is loose.
    d. Mixed results requiring discussion (MIXED):
        i. n/a

6. **Accepts inputs on the barcode scanner**.
    a. Procedure:
        i. Scan the first and second configuration barcode listed in the internal document *VxMark V4 - SW<>HW Integration*). Listen for a beep.
    b. Acceptable results (PASS):
        i. Scanner beeps.
    c. Unacceptable results (FAIL):
        i. No response from the scanner.
        ii. Scanner power is off; light is off.
    d. Mixed results requiring discussion (MIXED):
        i. n/a

Required Tests After Physical Stress Tests

Check the following before and after any environmental tests, vibrational tests, other mechanical tests, or any test that affects the corresponding parts (may not be needed for EMC tests).

7. **Maintains physical security**, restricting administrative access and preventing tampering without evidence.
    a. Procedure:
        i. Attempt to stuff ballots into the BB or MCM in a fully set up and secured system.
        ii. Inspect the poll worker door (administrative access door). Look for vulnerabilities in overcoming its locks and seals.
        iii. Look for exposed data ports (e.g. USB ports) when the poll worker door is closed.
    b. Acceptable results (PASS):
        i. No opportunity to stuff ballots into the BB or MCM.
        ii. No vulnerabilities seen in the pollworker door.
        iii. No exposed data ports seen when the poll worker door is closed.
    c. Unacceptable results (FAIL):
        i. Can stuff a ballot into the BB or MCM when fully set up and secured.
        ii. Broken poll worker door, or other door vulnerabilities found.
        iii. USB port accessible when poll worker door is closed/sealed.
    d. Mixed results requiring discussion (MIXED):

8. **Protects internal electronics and subsystems**.
    a. Procedure:
        i. Open the VxMark case and set it up as if for normal operation.
        ii. Inspect the screen, ATI, headphones, and main components.
        iii. Inspect the protective panels, fasteners, and case enclosing the internals. Look for any damage or openings.
        iv. Inspect the USB port, exposed cables, and cable storage.
        v. Close the VxMark case and inspect all sides.
        vi. Listen for loose parts internally.
        vii. If any damage is found, investigate further, potentially opening up the system to find more details.
    b. Acceptable results (PASS):
        i. No damage found.
    c. Unacceptable results (FAIL):
        i. Damage to protective panels/case that exposes internal electronics.
        ii. Damage to internal electronics found.
    d. Mixed results requiring discussion (MIXED):
        i. Cosmetic damage found.
        ii. Inconclusive wear or other damage found that could lead to bigger failures in future use cycles.

Inspection of Parts

Following the tests above, think about the following parts of the system. Inspect them if needed. Document if any problems below showed issues or damage:

1. Computer
2. Touchscreen
3. Power Supply, internal
4. Smart Card Reader
5. USB-A receptacle
6. USB 3 Switch
7. Barcode Scanner
8. Cable Assemblies
9. Uninterruptible Power Supply (external to unit)
10. Printer (external to unit)
11. Outer case
12. Inner panels around electronics
13. USB cover
14. Security tie points
15. Fasteners, both external and internal
16. Wires, ports, and electrical connections

Quick Tests, <1 minute

For doing very quick tests of normal function, for example between ESD gun tests, the following tests are recommended:

- Mark and print a valid ballot. Inspect by eye without scanning. (Test 1 above, but for a single valid ballot)

If no problems arise, that's a quick indicator that the system still works. If any problems arise, more in-depth analysis can be done.

Investigating Beyond What's in This Document

The key functions and components listed here cover much of the VotingWorks product requirements and government regulations for voting equipment (VVSG). However, **this document is not comprehensive** for indicating normal function. It only summarizes tests of normal function, requiring some tradeoffs to be done relatively quickly. If there is time to investigate more, and if there is a need to be more thorough, specific, or reliable, then please do so using engineering judgment.